Study notes: Complex Analysis I, Module 1, section 1.1

Zill and Shanahan, *Complex Analysis: A First Course with Applications* (3rd ed.), Section 1.1, printed pp. 3–9. Numbers are the book's own. No figures in this section. Exercises 1.1 are not included, except Problem 55, which Remark (i) names. Student notes in my own words; every definition, example and displayed equation is kept with its book number.

Before you start

- You can add, subtract and multiply real numbers, and you know that $(a+b)^2 = a^2 + 2ab + b^2$. Complex arithmetic uses the same rules, plus one new fact: $i^2 = -1$.
- Two real numbers are equal only when both their parts agree. For complex numbers this idea needs a formal statement (Definition 1.1.2 below).
- The real numbers can be ordered (you can say one is bigger than another). Complex numbers cannot. Remark (i) below explains this, and it is the first warning that complex analysis is not just real analysis with an extra symbol.

Section 1.1: Complex numbers and their properties

Why complex numbers were needed

Early algebra produced equations such as $x^2 + 2x + 2 = 0$ whose "solutions" contain $\sqrt{-1}$. Mathematicians resisted these numbers for centuries. The name "imaginary" comes from that resistance. In the eighteenth century Gauss gave the imaginary numbers a logical footing by treating them as an extension of the real numbers. This section sets out their definitions and arithmetic.

The imaginary unit and the complex number

The symbol i was first a disguise for $\sqrt{-1}$. Today we define i by the single property $i^2 = -1$. A complex number is built from two real numbers using i.

> **DEFINITION 1.1.1**
> A complex number is any number $z = a + ib$, where a and b are real numbers and i is the imaginary unit.

The notations $a + ib$ and $a + bi$ mean the same thing. The real number a is the **real part** of z, written $\mathrm{Re}(z)$. The real number b is the **imaginary part**, written $\mathrm{Im}(z)$. Note that $\mathrm{Im}(z)$ is a real number: for $z = 4 - 9i$, $\mathrm{Re}(z) = 4$ and $\mathrm{Im}(z) = -9$, not $-9i$. A real multiple of i, such as $6i$, is a **pure imaginary** number.

Equality

> **DEFINITION 1.1.2**
> Complex numbers $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$ are equal, $z_1 = z_2$, when $a_1 = a_2$ and $b_1 = b_2$.

Put in terms of Re and Im: $z_1 = z_2$ exactly when $\mathrm{Re}(z_1) = \mathrm{Re}(z_2)$ and $\mathrm{Im}(z_1) = \mathrm{Im}(z_2)$. Every real number a is the complex number $a + 0i$, so $\mathbb{R}$ sits inside $\mathbb{C}$ as a subset. The whole collection of complex numbers is written $\mathbb{C}$.

Addition, subtraction, multiplication and division

For $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$ the four operations are:

- **Addition:** $z_1 + z_2 = (a_1 + a_2) + i(b_1 + b_2)$.
- **Subtraction:** $z_1 - z_2 = (a_1 - a_2) + i(b_1 - b_2)$.
- **Multiplication:** $z_1 z_2 = a_1 a_2 - b_1 b_2 + i(b_1 a_2 + a_1 b_2)$.
- **Division:** $\dfrac{z_1}{z_2} = \dfrac{a_1 a_2 + b_1 b_2}{a_2^2 + b_2^2} + i\,\dfrac{b_1 a_2 - a_1 b_2}{a_2^2 + b_2^2}$, provided $z_2 \neq 0$ (that is, $a_2 \neq 0$ or $b_2 \neq 0$).

Commutative, associative and distributive laws all hold for complex numbers. So you do not need to memorise the formulas above. Two rules are enough: to add or subtract, work on real parts and imaginary parts separately; to multiply, expand with the distributive law and replace i^2 by -1.

Worked example: addition and multiplication

Example 1. Let $z_1 = 2 + 4i$ and $z_2 = -3 + 8i$. Then:

- (a) Add the real parts and the imaginary parts: $z_1 + z_2 = (2 - 3) + (4 + 8)i = -1 + 12i$.
- (b) Expand with the distributive law, one factor at a time, then use $i^2 = -1$:

$$z_1 z_2 = -6 - 12i + 16i + 32i^2 = -38 + 4i.$$

Zero and unity

The complex zero is $0 + 0i$ and the complex unity is $1 + 0i$. Zero does nothing when added: $z + 0 = (a + ib) + (0 + 0i) = a + ib = z$. Unity does nothing when multiplied: $z \cdot 1 = (a + ib)(1 + 0i) = a + ib = z$.

The conjugate

The **conjugate** $\bar{z}$ of $z = a + ib$ is $\bar{z} = a - ib$: flip the sign of the imaginary part. For example $\overline{6 + 3i} = 6 - 3i$ and $\overline{-5 - i} = -5 + i$. A real number is its own conjugate: $\bar{7} = 7$.

Conjugation respects addition and subtraction, and it also respects multiplication, division and taking the conjugate twice:

$$\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2, \qquad \overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2. \tag{1}$$

$$\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2, \qquad \overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}, \qquad \bar{\bar{z}} = z. \tag{2}$$

So the conjugate of any finite sum or product is the sum or product of the conjugates.

Two results from the definitions are used again and again:

$$z + \bar{z} = (a + ib) + (a - ib) = 2a. \tag{3}$$

$$z\bar{z} = (a + ib)(a - ib) = a^2 + b^2. \tag{4}$$

Equation (3) gives a real number. Equation (4) gives a real number too, and it is always ≥ 0. Equation (4) is the key to division, as the next heading shows.

The difference $z - \bar{z}$ is pure imaginary:

$$z - \bar{z} = (a + ib) - (a - ib) = 2ib. \tag{5}$$

Since $a = \operatorname{Re}(z)$ and $b = \operatorname{Im}(z)$, equations (3) and (5) give two formulas for the parts of z:

$$\operatorname{Re}(z) = \frac{z + \bar{z}}{2} \qquad \text{and} \qquad \operatorname{Im}(z) = \frac{z - \bar{z}}{2i}. \tag{6}$$

Division

To divide z_1 by z_2, multiply the top and bottom by $\bar{z}_2$. The bottom becomes $z_2\bar{z}_2$, which is real by (4), so the result is easy to write as $a + ib$:

$$\frac{z_1}{z_2} = \frac{z_1}{z_2} \cdot \frac{\bar{z}_2}{\bar{z}_2} = \frac{z_1 \bar{z}_2}{z_2 \bar{z}_2}. \tag{7}$$

Example 2. Let $z_1 = 2 - 3i$ and $z_2 = 4 + 6i$. Multiply by $\bar{z}_2 = 4 - 6i$ and use (4) on the bottom:

$$\frac{z_1}{z_2} = \frac{(2 - 3i)(4 - 6i)}{4^2 + 6^2} = \frac{-10 - 24i}{52} = -\frac{5}{26} - \frac{6}{13}i.$$

The last step divides the real and imaginary parts of the numerator by 52 and reduces each fraction.

Additive and multiplicative inverses

Every complex $z = a + ib$ has one additive inverse, $-z = -a - ib$, and $z + (-z) = 0$. Every *nonzero* complex z has exactly one multiplicative inverse z^{-1}, with $zz^{-1} = 1$. The inverse is the reciprocal $1/z$.

Example 3. Find the reciprocal of $z = 2 - 3i$. Multiply top and bottom by the conjugate $2 + 3i$:

$$\frac{1}{z} = \frac{1}{2 - 3i} \cdot \frac{2 + 3i}{2 + 3i} = \frac{2 + 3i}{4 + 9} = \frac{2}{13} + \frac{3}{13}i.$$

Check it yourself: $(2 - 3i)\left(\frac{2}{13} + \frac{3}{13}i\right) = 1$.

What is different from real analysis

Remarks. Two differences matter from the start.

i. **No order.** Real numbers can be compared with $<$. Complex numbers cannot, unless both are real. A statement such as $z_1 < z_2$ or $z_2 \geq z_1$ means nothing in $\mathbb{C}$ unless z_1 and z_2 are real. The book asks you to see why in Problem 55, which is printed in full below because Remark (i) names it. So when you see $z_1 = \alpha z_2$ with $\alpha > 0$, the inequality forces α to be real.

> **PROBLEM 55 (NAMED IN REMARK (I))**
> A number system is called an *ordered system* if it has a subset P with two properties. First, for any nonzero number x in the system, either x or $-x$ is in P, but not both. Second, if x and y are in P, then both xy and $x + y$ are in P. In the real numbers P is the set of positive numbers, and $x > y$ means $x - y \in P$. Discuss why the complex numbers have no such subset P. [Hint: consider i and $-i$.]

ii. **Impossible in the reals, ordinary in $\mathbb{C}$.** In real analysis $e^x = -2$ and $\sin x = 5$ have no solutions for real x. When x is a complex variable, both statements are correct and ordinary. See Example 3 in Section 4.1 and Example 2 in Section 4.3 of the book.

The book points out more differences throughout the text. Keep this list in mind when a later result looks odd.